



Koneru Lakshmaiah Education Foundation

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OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026–27, Term-I

Project Problem Statement Submission Form

Section / Team Details

Section No.: 03

Team No.:

Project Title: Linux Inter-Process Communication Using Pipes and FIFOs

Team Members

Roll Number	Student Name	Signature
2520030471	SAI ANANYA ROY	
2520030504	DEVUNI SHREENITHA	

1. Abstract

The Hospital Emergency Patient Monitoring System is a Linux-based project that demonstrates Inter-Process Communication (IPC) in a real-time hospital scenario. The system simulates patient monitoring, emergency processing, and alert generation using multiple processes. Patient data is transferred between processes using anonymous pipes and named pipes (FIFOs). Linux system calls such as `fork()`, `pipe()`, `mkfifo()`, `read()`, `write()`, and `wait()` are used for process creation, communication, and synchronization. The system is expected to efficiently transfer emergency information and generate alerts when a critical condition is detected.

2. Problem Statement

In hospitals, emergency patient information must be transferred quickly between monitoring and alert systems. This project demonstrates how multiple Linux processes can communicate and coordinate to process patient information using IPC mechanisms.

3. Objectives

- To simulate a hospital emergency monitoring system using Linux processes.

- To implement communication using pipes and FIFOs.
- To transfer and process patient emergency information.
- To demonstrate process creation, communication, and synchronization.

4. Proposed Methodology

The system consists of three processes: **Patient Monitoring, Emergency Processing, and Alert Process**. The monitoring process generates patient data and sends it through an anonymous pipe. The processing process checks the data for emergency conditions and sends the result through a FIFO. The alert process receives the message and displays an emergency alert.

5. Operating Systems Concepts / Linux APIs Used

OS Concept / Linux API / System Call	Purpose in the Project
fork()	Creates processes
pipe()	Communication between related processes
mkfifo()	Creates a named pipe
read()	Receives data
write()	Sends data
open() / close()	Manages communication channels
wait()	Process synchronization
IPC	Enables process communication
Process Synchronization	Coordinates processes
File Descriptors	Accesses pipes and FIFOs

6. Individual Contribution

Roll Number	Student Name	Individual Responsibility
2520030471	SAI ANANYA ROY	Process creation using fork(), unnamed pipe implementation using pipe(), data transfer using read()/write(), synchronization using wait(), testing and documentation
2520030504	DEVUNI SHREENITHA	Process creation using fork(), unnamed pipe implementation using pipe(), data transfer using read()/write(), synchronization using wait(), testing and documentation

7. Tools / Platforms / Software Used

Tool / Platform / Software	Purpose
Linux / Ubuntu	Project execution environment
C / C++	Program implementation
GCC	Compilation
POSIX APIs	System and IPC functions
VS Code/Nano	Compilation and testing
Linux Terminal	Code development

8. Expected Outcome

The system will simulate the transfer of patient information between multiple processes. It will detect critical conditions and generate an alert using IPC mechanisms. The project will demonstrate practical usage of Linux processes, pipes, FIFOs, and process synchronization in a hospital environment.

9. GitHub Repository

GitHub Repository Link: https://github.com/shreenithadevuni/Linux-Inter-Process-Communication-Using-Pipes-and-FIFOs-_OSSP

Repository Created: Yes

☐ Yes ☐ No

Faculty Name and Signature

Faculty Name: _____

Signature: _____

Remarks: _____

Faculty Use

Project Approved: _____

☐ Yes ☐ No

Date: _____

Faculty Signature: _____